



ABES ENGINEERING COLLEGE, GHAZIABAD (AUTONOMOUS)

Mid Term-I (Session: 2025-26)

Roll No.:

[Redacted Roll Number]

Course Name: B.Tech
Subject Name: Essentials of AI
Subject Code: 25AI101

Max. Marks: 15
Time: 75 Minutes
Semester: I

Note:

1. Keeping Mobile Phone/Any Electronic Gadgets, Even in OFF position is treated as UFM
2. Don't Write Anything on Question Paper
3. Attempt All Questions
4. Assume Any Missing Data if required

SECTION-A (Maximum Knowledge Level: K2)				
Q. No.	Question	CO	KL	PI
1. Attempt All Questions (1 x 5)				
(a)	Define supervised learning and unsupervised learning in AI.	CO1	K1	1.1.1
(b)	Define artificial intelligence in contrast to human intelligence.	CO1	K1	1.3.1
(c)	List at least two ethical considerations in AI development.	CO1	K1	5.3.1
(d)	State the characteristics of intelligent agents.	CO2	K1	2.1.1
(e)	List the four components of a PEAS description.	CO2	K1	3.1.1
SECTION-B (Minimum Knowledge Level: K2)				
2. Attempt Any two Question (Same K Levels) (2 x 2)				
(a)	Explain the relationship among AI, Machine Learning (ML), and Deep Learning (DL).	CO1	K2	1.4.1
(b)	Explain the differences between supervised, unsupervised, and reinforcement learning.	CO1	K2	1.1.2
(c)	Differentiate between discrete and continuous probability distributions with examples.	CO1	K2	2.4.1
SECTION-C (Minimum Knowledge Level: K3)				
3. Attempt Any one Question (Same K Levels) (3 x 1)				
(a)	Use the concept of reinforcement learning to describe how a self-driving car could learn to navigate a city.	CO1	K2	2.2.3
(b)	Compute the mean, median, and mode for the dataset: 5, 7, 9, 7, 10, 12.	CO1	K2	2.1.3
4. Attempt Any one Question (Same K Levels) (3 x 1)				
(a)	Illustrate the decision-making process of a goal-based agent.	CO2	K2	4.1.2
(b)	Apply the PEAS framework to design an AI agent for a smart home environment.	CO2	K2	3.2.1

CO Course Outcomes mapped with respective question

KL Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create



ABES ENGINEERING COLLEGE, GHAZIABAD (AUTONOMOUS)

Mid Term-II (Session: 2025-26)

Roll No.:

2

Course Name: B.Tech
Subject Name: Essentials of AI
Subject Code: 25AI101

Max. Marks: 15
Time: 75 Minutes
Semester: I

Note:

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SECTION-A (Maximum Knowledge Level: K2)				
Q. No.	Question	CO	KL	PI
1. Attempt All Questions (1 x 5)				
(a)	What are the key characteristics of the Python programming language?	CO3	K2	1.3.1
(b)	Write a Python loop to print even numbers between 1 and 10.	CO3	K2	2.4.1
(c)	Write a simple Python program to display "Hello, World!".	CO3	K2	1.3.1
(d)	Differentiate between propositional logic and predicate logic.	CO2	K2	1.4.1
(e)	Discuss types of problems in AI.	CO2	K2	1.1.1
SECTION-B (Minimum Knowledge Level: K2)				
2. Attempt Any two Question (Same K Levels) (2 x 2)				
(a)	Explain the various types of python packages.	CO3	K2	1.4.1
(b)	Describe the various types of operators in Python. Provide examples of arithmetic, logical, relational, assignment, and bitwise operators.	CO3	K2	1.3.1
(c)	Write a Python program to check whether a given number is prime using a user-defined function.	CO3	K2	2.4.1
SECTION-C (Minimum Knowledge Level: K3)				
3. Attempt Any one Question (Same K Levels) (3 x 1)				
(a)	Write the output of the code- total = 0 for i in range(1, 11): total += i print("Sum of first 10 natural numbers:", total)	CO3	K3	2.4.1
(b)	(i) Discuss Datatypes in python give one example of each. (ii) Write the output of the code- my_list = [10, 20, 30, 40, 50] for item in my_list: print(item)	CO3	K3	1.3.1
4. Attempt Any one Question (Same K Levels) (3 x 1)				
(a)	Describe AI Knowledge Cycle with its components.	CO2	K2	2.1.1
(b)	What are the types of search strategies used in AI problem-solving? Discuss in detail the difference between uninformed and informed search algorithms with examples.	CO2	K2	2.1.3

CO Course Outcomes mapped with respective question
KL Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create



Admission No.:

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Semester: I

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*CO Course Outcomes mapped with respective question
KL Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)
K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create*